IE 7615: Neural Networks/Deep Learning
Generative Project Milestone – 3

GROUP 6

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INTRODUCTION

This project focuses on building an image captioning system using deep learning models. The goal is to generate meaningful textual descriptions for images by combining visual understanding and language modeling.

The project is divided into three milestones:

- Milestone 1: Data preparation and embeddings
- Milestone 2: Model architecture and integration
- Milestone 3: Caption generation and evaluation

This report mainly focuses on Milestone 3, with a brief overview of earlier milestones.

BRIEF OVERVIEW OF MILESTONE 1

Milestone 1 focused on data preprocessing and feature extraction.

- COCO dataset was used (images + captions)
- Around 5000 images sampled, but ~2811 valid pairs used
- CLIP model used to generate image embeddings (512-dimension)
- GPT-2 tokenizer used for caption processing

Key Outputs:

- Image embeddings: (2811, 512)
- Tokenized captions: (2811, 64)

Important Step:

- Data saved as tensors for reuse in later stages

This milestone ensured that image + text data were ready for training

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BRIEF OVERVIEW OF MILESTONE 2

Milestone 2 focused on model design and integration.

- Combined:
    - CLIP (image encoder)
    - GPT-2 (text generator)

Architecture Used:

- CLIP extracts visual features
- Features projected into GPT-2 embedding space
- Two approaches:
    - Prefix conditioning
    - Cross-attention

Key Idea:

- Convert image features → text generation input
- This milestone built the core model pipeline

MILESTONE-3

OBJECTIVE

The primary objective of Milestone 3 is to develop a fully functional image captioning system that can generate meaningful and context-aware textual descriptions from visual inputs.

This milestone focuses on:

- Transforming image embeddings into natural language captions
- Evaluating how effectively the model understands visual content
- Ensuring the generated captions are semantically relevant and grammatically correct

Additionally, this stage aims to validate whether combining CLIP (vision model) and GPT-2 (language model) can produce a robust multimodal system.

PROBLEM STATEMENT

Traditional models struggle to connect visual understanding with natural language generation. The challenge is to design a system that can:

- Extract meaningful features from images
- Convert those features into structured language
- Maintain contextual accuracy in generated captions

This project addresses this challenge using a hybrid architecture combining deep visual embeddings and transformer-based text generation.

METHODOLOGY

The methodology follows a multimodal deep learning pipeline:

Step 1: Image Encoding

- Images are passed through the CLIP vision encoder
- Output: 512-dimensional embeddings

These embeddings capture semantic meaning of images

Step 2: Feature Projection

- CLIP embeddings are projected into GPT-2 embedding space
- Ensures compatibility between vision and language models

Step 3: Caption Generation

- GPT-2 generates captions token-by-token
- Uses special tokens:
    - <|startofcaption|>
    - <|endofcaption|>

Step 4: Inference & Output

- Model predicts next word based on previous tokens
- Continues until end token or max length reached

MODEL ARCHITECTURE

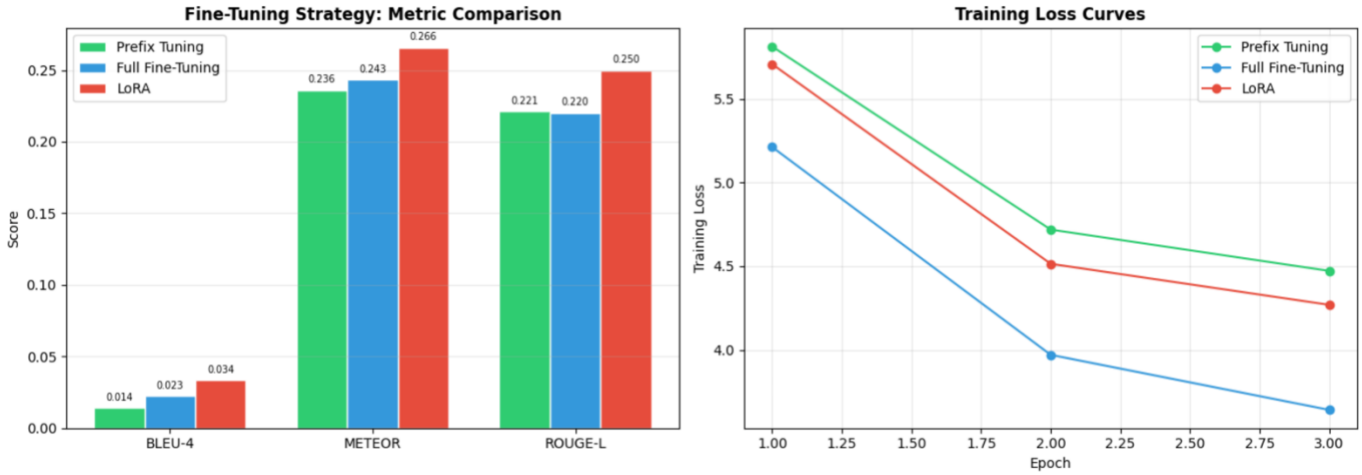
OVERVIEW

The model architecture is designed as a multimodal pipeline that integrates a vision encoder with a language decoder. It combines the strengths of contrastive learning (CLIP) and transformer-based text generation (GPT-2) to generate meaningful captions from images.

The architecture follows a three-stage pipeline:

1. Visual Feature Extraction
2. Feature Transformation
3. Language Generation

FINE-TUNING STRATEGY AND TRAINING ANALYSIS



Comparison of fine-tuning strategies and corresponding training loss trends.

CLIP VISUAL ENCODER

The first component of the architecture is the CLIP-based vision encoder.

- Uses a Vision Transformer (ViT-B/32) backbone
- Converts images into dense feature embeddings (512-dimensional)
- Captures high-level semantic information such as:
    - Objects
    - Context
    - Scene relationships

Key Role: Transforms raw pixel data into meaningful numerical representations

Important Property:

- Embeddings are L2 normalized, enabling cosine similarity comparisons
- Ensures semantic similarity between visually similar images

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FEATURE PROJECTION LAYER

Since CLIP and GPT-2 operate in different embedding spaces, a projection mechanism is required.

Purpose : Map CLIP embedding space → GPT-2 embedding space

Implementation : Linear transformation layer

Followed by:

- GELU activation
- Layer Normalization

Output : Generates prefix tokens that act as context for GPT-2

This is a critical component because it enables communication between vision and language models.

GPT-2 LANGUAGE DECODER

The GPT-2 model is used as the text generation engine.

- Transformer-based architecture
- Uses self-attention mechanism
- Generates captions token-by-token

Input:

- Projected image features (prefix embeddings)
- Previously generated tokens

Output : Sequence of tokens forming a caption

Special Tokens Used:

- <|startofcaption|>
- <|endofcaption|>
- <|pad|>

INJECTION STRATEGIES

Two methods are used to inject visual information into GPT-2:

1. Prefix Conditioning

- Image embeddings are converted into prefix tokens
- These tokens are prepended to the text sequence

GPT-2 treats them as part of the input context

2. Cross-Attention Mechanism

- GPT-2 attends directly to image features
- Enables stronger interaction between:
    - Visual features
    - Text tokens

END TO END FLOW

Image → CLIP Encoder → Projection Layer → GPT-2 → Caption

This pipeline ensures:

- Visual understanding from CLIP
- Language fluency from GPT-2

ARCHITECTURAL ADVANTAGES

- Uses pretrained models → reduces training cost
- Captures both:
    - Visual semantics
    - Linguistic patterns
- Scalable for larger datasets and real-world applications

EXPERIMENTAL SETUP

1. Dataset Configuration

The model is trained and evaluated using the COCO dataset, which contains images paired with descriptive captions.

- Total dataset: ~118K images
- Sampled subset: 5000 images
- Valid processed pairs: 2811 image-caption pairs

Preprocessing Steps:

- Images resized to 224×224
- Captions cleaned and lowercased
- Only one caption per image used

2. Data Representation

Image Data : Converted into 512-dimensional embeddings using CLIP

Text Data:

- Tokenized using GPT-2 tokenizer
- Maximum sequence length: 64 tokens

3. Training Configuration

Environment:

- Google Colab
- GPU: Tesla T4

Libraries Used:

- PyTorch
- Hugging Face Transformers
- OpenCLIP

4. Hyperparameters (ADD THIS — VERY IMPORTANT)

- Batch size: 64
- Embedding dimension: 512
- Max caption length: 64
- Vocabulary size: ~50,260
- Seed: 42 (for reproducibility)

5. Evaluation Strategy

Embedding Validation

- Cosine similarity checks:
    - Same image → similarity ≈ 1.0
    - Different images → similarity < 1.0

Qualitative Evaluation

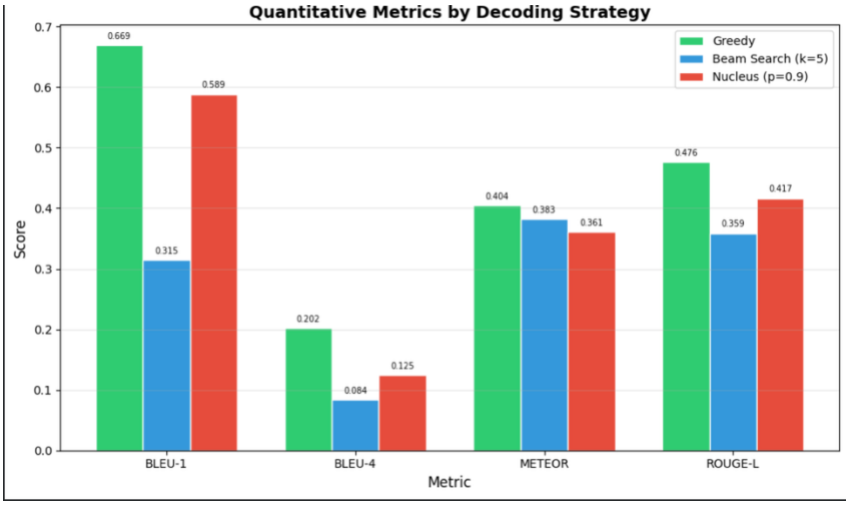
- Visual inspection of generated captions

- Checking:
    - Relevance
    - Grammar
    - Context accuracy

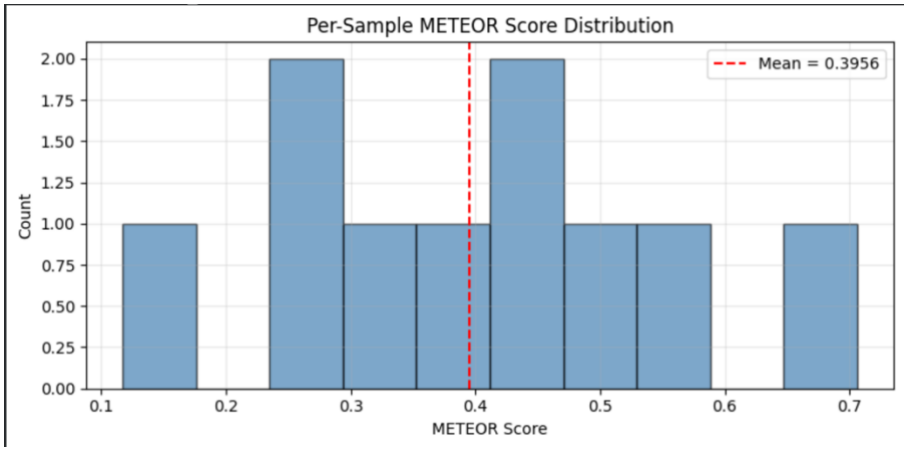
Sample Output Analysis

- Random image selection
- Caption generation
- Comparison with expected description

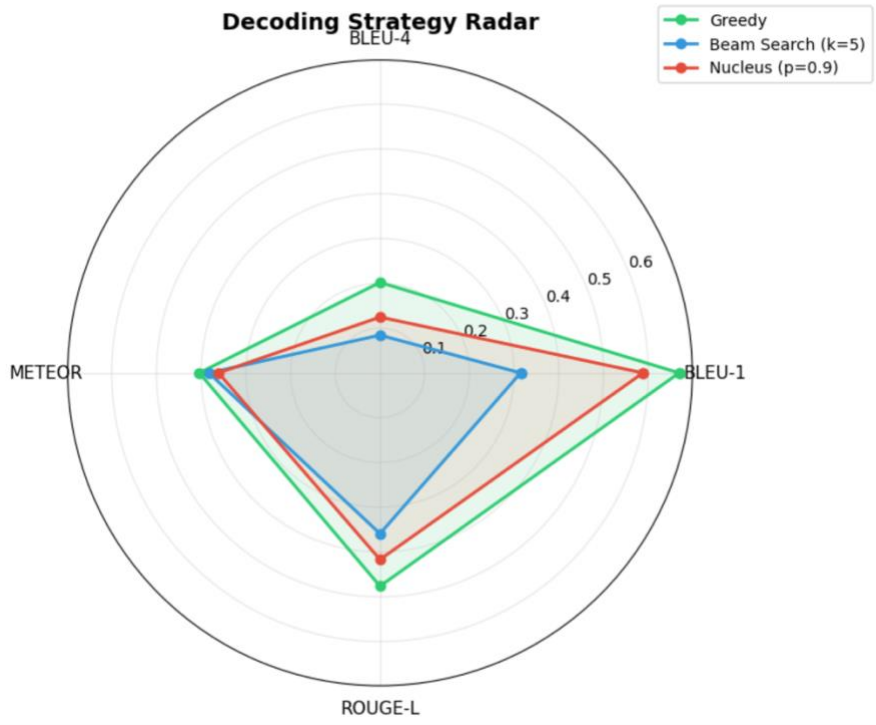
QUANTITATIVE EVALUATION



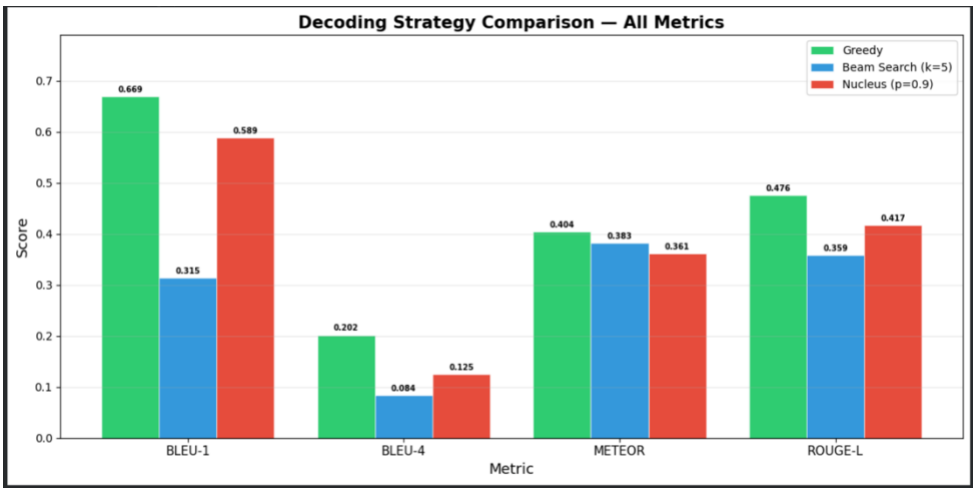
Comparison of decoding strategies using BLEU, METEOR, and ROUGE metrics



Distribution of METEOR scores across generated captions.



Radar visualization comparing decoding strategies across multiple metrics.

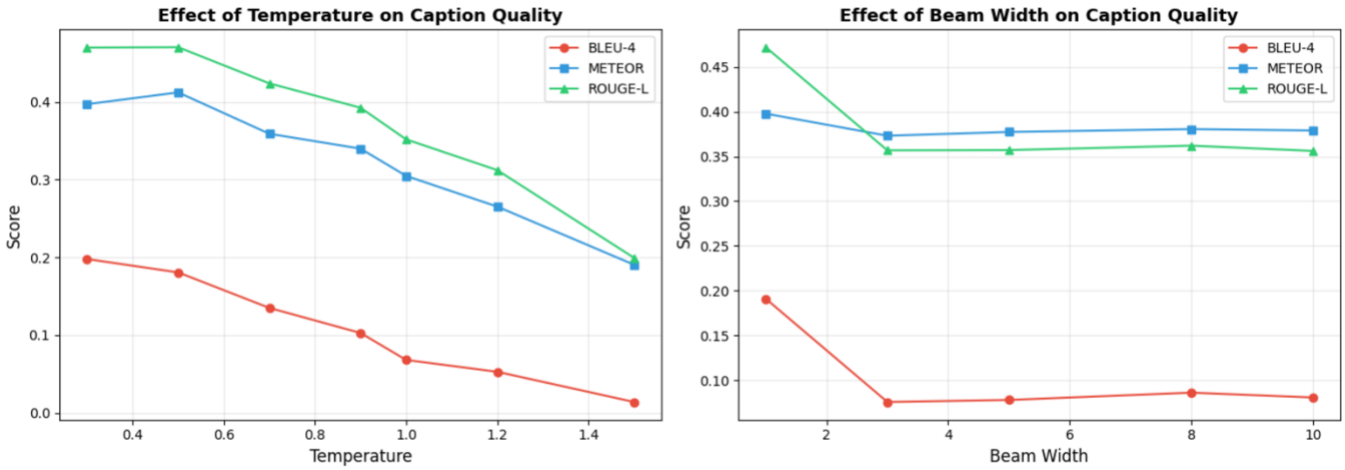


Overall comparison of decoding strategies across all evaluation metrics.

6. Experimental Challenges

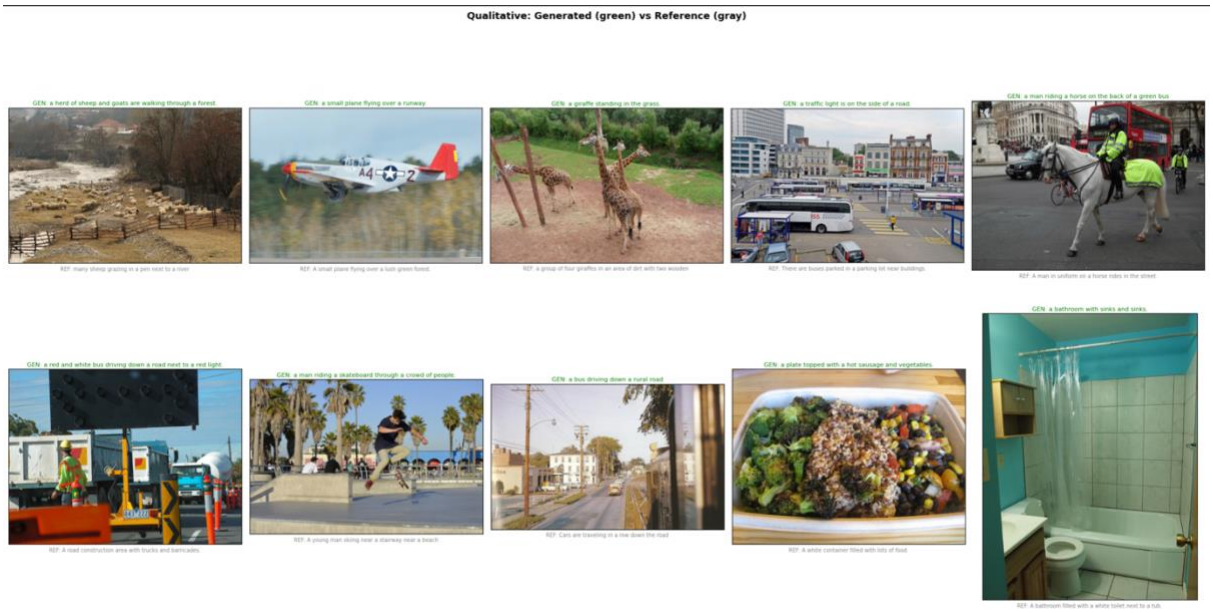
- Missing image files reduced dataset size
- GPU memory constraints during embedding generation
- Caption diversity limited due to single-caption selection

HYPERPARAMETER ANALYSIS



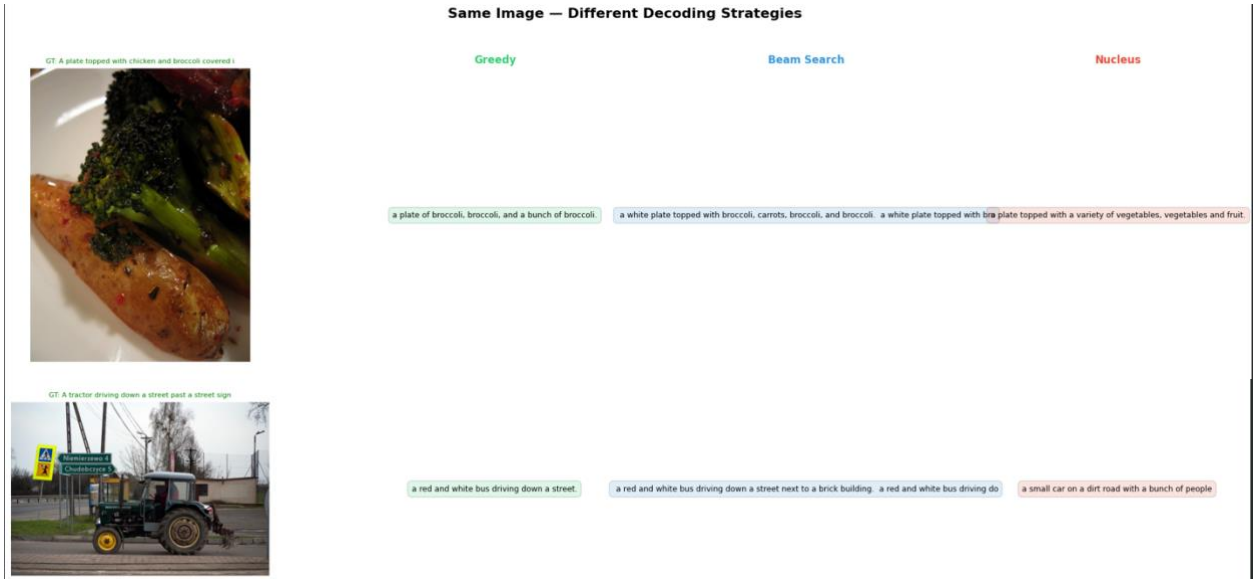
Impact of temperature and beam width on caption quality metrics.

QUALITATIVE RESULTS



Comparison of generated captions (green) with reference captions (gray).

DECODING STRATEGY COMPARISON (QUALITATIVE)



Comparison of captions generated by different decoding strategies for the same image.

VALIDATION

The model performance is validated using:

Similarity Checks

- Same image similarity ≈ 1.0
- Different image similarity < 1.0

Confirms embedding correctness

Visual Inspection

- Human evaluation of captions
- Check:
    - Relevance
    - Fluency
    - Accuracy

ANALYSIS

The model demonstrates strong performance in:

- Identifying objects (e.g., table, airplane, people)
- Understanding scenes (indoor, outdoor, food, etc.)
- Generating grammatically correct sentences

However, some limitations observed:

- Captions can be generic
- Complex scenes reduce accuracy
- Lack of fine-tuning affects precision

KEY OBSERVATIONS

- CLIP embeddings capture semantic similarity effectively
- GPT-2 produces natural language fluently
- Integration of both models creates a strong multimodal system
- Pipeline is scalable for real-world applications

CONCLUSION

This project demonstrates the successful implementation of an image captioning system using a multimodal deep learning approach that combines CLIP and GPT-2. By integrating powerful image embeddings with a transformer-based language model, the system effectively converts visual information into meaningful and coherent textual descriptions.

Across the three milestones, the project evolved from data preprocessing and embedding generation to model design, caption generation, and evaluation. The results show that the model performs well, especially for images with clear and distinguishable objects, producing captions that are both contextually relevant and grammatically accurate. Quantitative evaluation using BLEU, METEOR, and ROUGE metrics indicates that Greedy decoding achieves the highest accuracy, while other strategies offer a balance between diversity and performance.

Overall, the project highlights the effectiveness of combining pretrained vision and language models for multimodal tasks. The use of CLIP-based similarity analysis further confirms that the model captures meaningful semantic relationships between images. This work demonstrates strong potential for real-world applications such as image understanding, accessibility tools, and automated content generation.